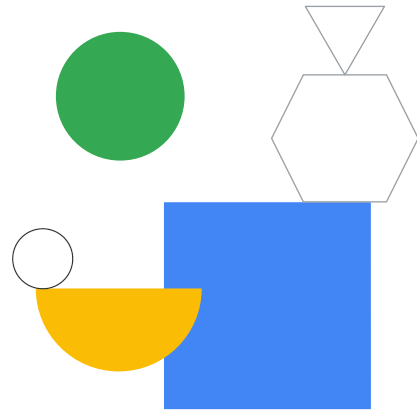


Preparing for Your Associate Cloud Engineer Journey

Section 4: Ensuring Successful Operation of a Cloud Solution



Welcome to section 4 of the exam guide: Ensuring Successful Operation of a Cloud Solution.



Module agenda

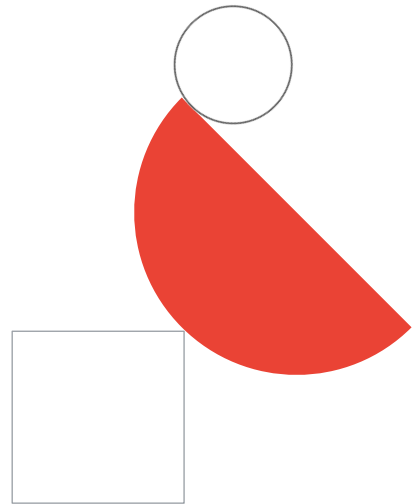
- 01 Managing Cymbal Superstore's cloud solutions
- 02 Diagnostic questions
- 03 Review and study planning

In this module you'll explore the scope of ensuring successful cloud operations. This involves managing the compute, storage, and networking resources as well as monitoring and logging tasks. These areas correspond to the fourth section of the Associate Cloud Engineer Exam Guide.

We'll start by discussing your role as an Associate Cloud Engineer in managing Cymbal Superstore's cloud solutions. Next, you'll assess your skills in this area through 9 diagnostic questions.

When we review the questions, identify the resources you'll want to include in your study plan.

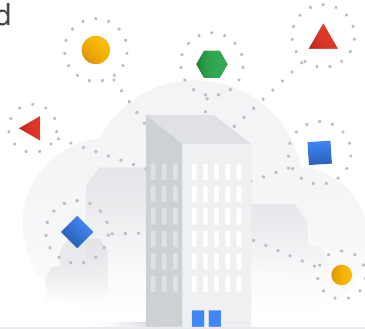
Managing Cymbal Superstore's cloud solutions



Now that Cymbal Superstore's cloud solutions have been deployed and implemented, your role as an Associate Cloud Engineer shifts focus to maintaining successful operations. Let's explore what that means.

The next step:

managing Cymbal
Superstore's cloud
solutions



- Managing Compute Engine resources
- Managing GKE resources
- Managing Cloud Run resources
- Managing storage and database solutions
- Managing networking resources
- Monitoring and logging



In deploying and implementing Cymbal Superstore's cloud architecture, you needed to know how to work with various compute, storage, and networking resources on Google Cloud. To ensure successful operations, an Associate Cloud Engineer needs the knowledge and skills to manage the resources used in an organization's cloud solutions. You also need to be able to use Google Cloud Observability for monitoring and logging.

Managing Cymbal Superstore's **supply chain app**

Upgrading managed instance groups:

```
gcloud compute instance-groups managed
rolling-action start_update cymball_supplychain_ig \
  --version=template=cymball_supplychain_ig_templat
e_<yymmdd> \
  --type=proactive\
  --region=us-central1
```

Google Cloud

Cymbal Superstore's supply chain management app is made of resources implemented close to their HQ in Minneapolis, Mn. It is architected using Compute Engine. Managed instance groups let the application scale automatically and remain available across zonal outages.

Sometimes the instance template that the group is based on might need to be changed.

Some reasons you might want to do this include the following:

- Updating the operating system of your instances.
- Conducting A/B or canary testing of capability upgrades.
- Changing the disk type or attached disks attached to your instances.

Once you do update the template, you need to ensure the change is propagated to all the VM instances in the group.

Managing Cymbal Superstore's **ecommerce app**

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: cymbal-ecommerce-ingress
  annotations:
    # If the class annotation is not specified
    # it defaults to "gce".
    kubernetes.io/ingress.class: "gce"
spec:
  rules:
    - http:
        paths:
          - path: /sales
            pathType: ImplementationSpecific
            backend:
              service:
                name: sales-service
                port:
                  number: 60000
```

```
- path: /support
  pathType: ImplementationSpecific
  backend:
    service:
      name: support-service
      port:
        number: 80
```

Here is an example of an ingress object that implements an external layer 7 (http(s)) load balancer

Cymbal Superstore's ecommerce app is architected using containers deployed to GKE pods. As an Associate Cloud Engineer on the ecommerce team, you might be asked to configure and monitor external connectivity. An external http(s) load balancer is a solution that advertises a single global IP, provides content close to your end user, and forwards content to backends that are available globally.

Managing Cymbal Superstore's transportation management app

Querying external data such as Bigtable data with BigQuery:

1. create a table definition file
2. create a permanent external table in BigQuery

```
bq mk --external_table_definition=cymbal_trans_mngt_bt_def /  
    cymbal_data_set.trans_mngt_ext_tbl
```

3. Query the data using the permanent table reference in the from clause of a sql query

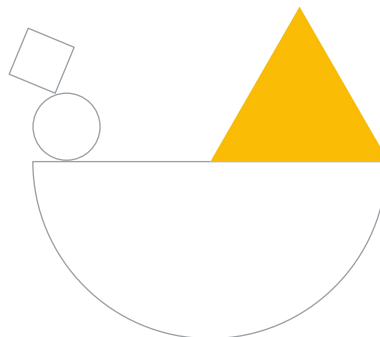
Google Cloud

Cymbal Superstore's transportation management app uses Cloud Run functions to monitor incoming sensor data and implements a Dataflow pipeline that uses a sink to write data to Bigtable. As an Associate Cloud Engineer, it is common to provide information about the sources and sinks required by a pipeline to the data engineer responsible for developing it. You also need to know how to monitor your incoming data stream and manage your Cloud Run function instances.

Let's specifically discuss how you might set up the resources required to query that data on a regular basis.

What would this involve? Let's think about Cymbal Superstore's transportation management app. You can use BigQuery sql to query your Bigtable data by defining a permanent external table using the Google Cloud Console or the bq command line tool. You do this by creating a table definition file which includes the uri for the table in Bigtable and information about the column families and columns defined in the table. The entries for the table definition are written in JSON. Next you create an external table reference with a bq mk command. Here is an example of what that might look like for our Cymbal Superstore example. Then you can submit the query using BigQuery sql.

Diagnostic questions



Now that you have some context for the objectives in this exam section, it's time to take a self-assessment focused on managing Google Cloud solutions.

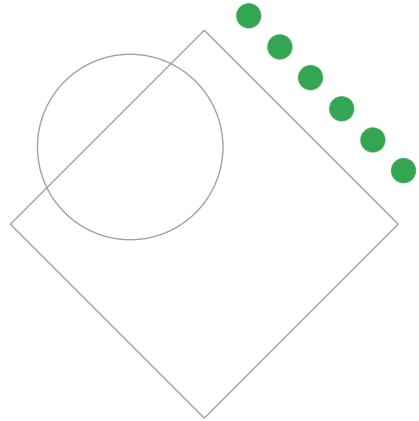
Please complete the diagnostic questions now

- The diagnostic questions are available in the workbook.



Please take 15 minutes to complete the diagnostic questions for this section.

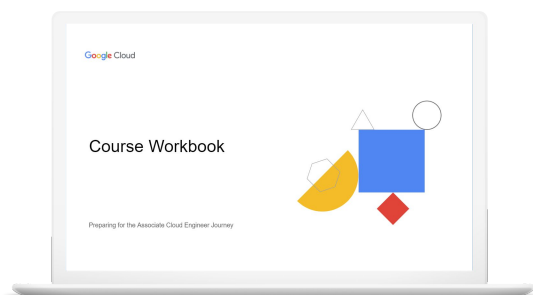
Review and study planning



What areas do you need to develop your skills in order to manage the different aspects of a Google Cloud solution? This is another important area for an Associate Cloud Engineer, and where you'll likely spend much of your time on the job. Let's review the diagnostic questions to help you target your study time to focus on the areas where you need to develop your skills.

Your study plan:

Ensuring successful operation of a cloud solution



- 4.1 Managing Compute Engine resources
- 4.2 Managing Google Kubernetes Engine resources
- 4.3 Managing Cloud Run resources
- 4.4 Managing storage and database solutions
- 4.5 Managing networking resources
- 4.6 Monitoring and logging

We'll approach this review by looking at the objectives of this exam section and the questions you just answered about each one. We'll introduce an objective, briefly review the answers to the related questions, then talk about where you can find out more in the learning resources and/or in Google Cloud documentation. As we go through each section objective, use the page in your workbook to mark the specific documentation, courses (and modules!), and skill badges you'll want to emphasize in your study plan.

Just like with the previous section, there are multiple objectives in this section that have many related tasks - so you will probably need to plan for more study time.

4.1 | Managing Compute Engine resources

Considerations include:

- Remotely connecting to the instance
- Viewing current running VM inventory (e.g., instance IDs, details)
- Working with snapshots (e.g., create a snapshot from a VM, view snapshots, delete a snapshot, schedule a snapshot)
- Working with images (e.g., create an image from a VM or a snapshot, view images, delete an image)

As we've discussed before, Cymbal Superstore's supply chain app is built on Compute Engine resources.

Working with snapshots and images is a crucial part of managing Compute Engine resources. As an Associate Cloud Engineer, you might need to use these tools to create backups of critical data, restore VMs to a previous state in case of a failure or misconfiguration, or to deploy new VMs quickly and consistently using pre-configured images.

These are the diagnostic questions you answered that relate to this area:

Question 1: Identify commands required to list and describe Compute Engine disk snapshots

Question 2: Describe the incremental nature of Compute Engine disk snapshots

4.1 | Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. gcloud compute snapshots list
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots

Question:

You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

4.1 Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. **gcloud compute snapshots list**
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots



Google Cloud

Feedback:

*A. **gcloud compute snapshots list**

Feedback: Correct! gcloud commands are built with groups and subgroups, followed by a command, which is a verb. In this example, Compute is the Group, snapshots is the subgroup, and list is the command.

B. **gcloud snapshots list**

Feedback: Incorrect. Snapshots is not a main group defined in gcloud.

C. **gcloud compute snapshots get**

Feedback: Incorrect. Available commands for snapshots are list, describe, and delete.

D. **gcloud compute list snapshots**

Feedback: Incorrect. Snapshots is a compute command subgroup. It needs to come before the list command.

Where to look:

<https://cloud.google.com/compute/docs/disks/create-snapshots#listing-snapshots>
<https://cloud.google.com/compute/docs/disks/create-snapshots#viewing-snapshot>

Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M3 Virtual Machines and Networks in the Cloud

- Architecting with Google Compute Engine (ILT)
 - M3 Virtual Machines
- Essential Google Cloud Infrastructure: Foundation (On-demand)
 - M3 Virtual Machines

Summary:

Explanation/summary on the following slide.

Getting information about snapshots

To list Compute Engine disk snapshots:

```
gcloud compute snapshots list --project PROJECT_ID
```

To describe snapshots:

```
gcloud compute snapshots describe SNAPSHOT_NAME
```

To list snapshots run the following command:

```
gcloud compute snapshots list --project PROJECT_ID
```

Flags available

```
--limit maximum number of results you want back  
--regexp a filter for results you want returned  
--sort-by field to sort by  
--uri only print URI's of the resources returned
```

To describe snapshots, which returns creation time, size, and source disk, run the following command:

```
gcloud compute snapshots describe SNAPSHOT_NAME
```

Flags available

```
--format what kind of format you want printed out  
--project project to be used for command  
--quiet disables interactive prompts
```


4.1 | Diagnostic Question 02 Discussion



You have a scheduled snapshot you are trying to delete, but the operation returns an error.

What should you do to resolve this problem?

- A. Delete the downstream incremental snapshots before deleting the main reference.
- B. Delete the object the snapshot was created from.
- C. Detach the snapshot schedule before deleting it.
- D. Restore the snapshot to a persistent disk before deleting it.

Question:

You have a scheduled snapshot you are trying to delete, but the operation returns an error. What should you do to resolve this problem?

4.1 | Diagnostic Question 02 Discussion



You have a scheduled snapshot you are trying to delete, but the operation returns an error.

What should you do to resolve this problem?

- A. Delete the downstream incremental snapshots before deleting the main reference.
- B. Delete the object the snapshot was created from.
- C. Detach the snapshot schedule before deleting it. ✓
- D. Restore the snapshot to a persistent disk before deleting it.

Google Cloud

Feedback:

A. Delete the downstream incremental snapshots before deleting the main reference.
Feedback: Incorrect. This is not required to delete a scheduled snapshot and would be a lot of manual work.

B. Delete the object the snapshot was created from.

Feedback: Incorrect. This is not required to delete a scheduled snapshot and is destructive.

*C. Detach the snapshot schedule before deleting it.

Feedback: Correct! You can't delete a snapshot schedule that is still attached to a persistent disk.

D. Restore the snapshot to a persistent disk before deleting it.

Feedback: Incorrect. This does not allow you to delete a scheduled snapshot.

Where to look:

<https://cloud.google.com/compute/docs/disks/snapshots#incremental-snapshots>

Content mapping:

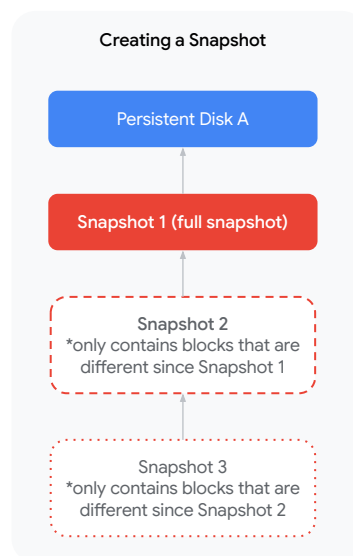
- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M3 Virtual Machines and Networks in the Cloud
- Architecting with Google Compute Engine (ILT)

- M3 Virtual Machines
- Essential Google Cloud Infrastructure: Foundation (On-demand)
 - M3 Virtual Machines

Summary:

Explanation/summary on the following slide.

Snapshots are incremental



Reference from snapshot concepts page:

<https://cloud.google.com/compute/docs/disks/snapshots>

Snapshots are incremental in nature and less expensive than creating full images of a disk. You can only create them for persistent disks. They are stored in a Cloud Storage bucket managed by the snapshot service and are automatically compressed. You can choose regional or multi-regional storage, which will affect cost. Snapshots are stored across multiple locations with automatic checksums. You schedule them using the gcloud command line and cron. You can also set up a snapshot schedule in the Google cloud console or command line. A Snapshot schedule and its source persistent disk have to be in the same region. You can restore them to a new persistent disk. The new disk can be in a different zone or region, so you can use snapshots to move VMs. You can create them while they are running. Snapshots of a disk have to be at least 10 minutes apart.

Here's how the incremental nature of snapshots works:

- The first snapshot is full and contains all data on the persistent disk it was run on.
- Each subsequent snapshot only contains new or modified data since the first snapshot.
- However, sometimes to clean up resources and cost, a new snapshot might be a full backup.

Deleting a snapshot copies its data to downstream incremental snapshots that are dependent on it, increasing the downstream snapshot's size. You can't delete a snapshot that has a schedule associated with it.

4.1 Managing Compute Engine resources

Courses



[Google Cloud Fundamentals: Core Infrastructure](#)

- M3 Virtual Machines and Networks in the Cloud

[Architecting with Google Compute Engine](#)

- M3 Virtual Machines
- M9 Load Balancing and Autoscaling



=

[Essential Google Cloud Infrastructure: Foundation](#)

- M3 Virtual Machines

[Elastic Google Cloud Infrastructure: Scaling and Automation](#)

- M2 Load Balancing and Autoscaling



Documentation

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Persistent disk snapshots | Compute Engine Documentation](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Foundation \(On-demand\)](#)

[Elastic Google Cloud Infrastructure: Scaling and Automation \(On-demand\)](#)

<https://cloud.google.com/compute/docs/disks/create-snapshots#listing-snapshots>

<https://cloud.google.com/compute/docs/disks/create-snapshots#viewing-snapshot>

<https://cloud.google.com/compute/docs/disks/snapshots#incremental-snapshots>

4.2 | Managing Google Kubernetes Engine resources

Considerations include:

- Viewing current running cluster inventory (e.g., nodes, pods, services)
- Configuring Google Kubernetes Engine to access Artifact Registry
- Working with node pools (e.g., add, edit, or remove a node pool)
- Working with Kubernetes resources (e.g., Pods, Services, Statefulsets)
- Managing Horizontal and Vertical autoscaling configurations.

Google Cloud

GKE is a managed service that provides production-level orchestration for your container-based applications. Cymbal Superstore's e-commerce app is implemented through services being exposed by GKE. As an Associate Cloud Engineer on the ecommerce app team, certain tasks might require you to interact with a GKE cluster and its nodes. You will also have to know about GKE's workload objects, such as pods, deployments, and services. Containers in GKE are based on images which are shared via Artifact Registry. You need to be familiar with how to create images and deploy them to the registry.

These diagnostic questions addressed managing GKE resources:

Question 3: Contrast the differences between an internal and external load balancer in Google Kubernetes Engine

Question 4: Describe the relationship between Kubernetes pods, services, and deployments

Question 5: Apply kubectl commands to manage pods, deployments, and services

4.2 | Diagnostic Question 03 Discussion



Cymbal Superstore's GKE cluster requires an internal Application Load Balancer. You are creating the configuration files required for this resource.

What is the proper setting for this scenario?

- A. Annotate your ingress object with an `ingress.class` of "gce."
- B. Configure your service object with a type: `LoadBalancer`.
- C. Annotate your service object with a "neg" reference.
- D. Implement custom static routes in your VPC.

Question:

Cymbal Superstore's GKE cluster requires an internal Application Load Balancer. You are creating the configuration files required for this resource. What is the proper setting for this scenario?

4.2 Diagnostic Question 03 Discussion



Cymbal Superstore's GKE cluster requires an internal Application Load Balancer. You are creating the configuration files required for this resource.

What is the proper setting for this scenario?

- A. Annotate your ingress object with an ingress.class of "gce."
- B. Configure your service object with a type: LoadBalancer.
- C. Annotate your service object with a "neg" reference. ☒
- D. Implement custom static routes in your VPC.

Google Cloud

Feedback:

A. Annotate your ingress object with an ingress.class of "gce."

Feedback: Incorrect. To implement an internal load balancer, the ingress class needs to be "gce-internal."

B. Configure your service object with a type: LoadBalancer.

Feedback: incorrect. Using LoadBalancer at the service level implements a Layer 4 Network Load Balancer, not an Application Load Balancer.

*C. Annotate your service object with a "neg" reference.

Feedback: Correct! This is correct because an internal Application Load Balancer can only use NEG's.

D. Implement custom static routes in your VPC.

Feedback: Incorrect. This describes a routes-based cluster. In order to support internal load balancing, your cluster needs to use VPC-native mode, where your cluster provides IP addresses to your pods from an alias IP range.

Where to look:

<https://cloud.google.com/kubernetes-engine/docs/concepts/ingress-ilb>

<https://cloud.google.com/kubernetes-engine/docs/concepts/ingress-xlb>

<https://cloud.google.com/kubernetes-engine/docs/how-to/load-balance-ingress>

<https://cloud.google.com/kubernetes-engine/docs/how-to/internal-load-balance-ingress>

S

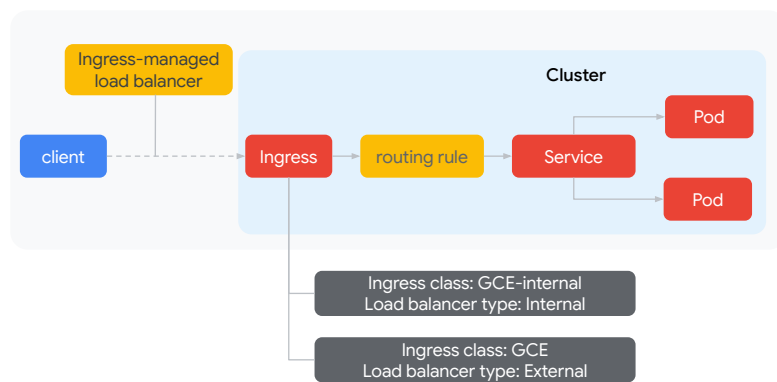
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M5 Containers in the Cloud
- Getting Started with Google Kubernetes Engine (ILT and On-demand)
 - M4 Kubernetes Operations

Summary:

Explanation/summary on the following slide.

Internal vs External load balancing in Kubernetes



To implement network load balancing you create a service object with these settings:

- `type: LoadBalancer`.
- Set External Traffic Policy to cluster or local

Cluster - traffic will be load balanced to any healthy GKE node and then kube-proxy will send it to a node with the pod.

Local - nodes without the pod will be reported as unhealthy. Traffic will only be sent to nodes with the pod. Traffic will be sent directly to pod with source ip header info included.

To implement an external Application Load Balancer, create an ingress object with the following settings:

- Routing depends on URL path, session affinity, and the balancing mode of backend Network endpoint groups (NEGS)
- The object type is ingress.
- Using `ingress.class: "gce"` annotation in the metadata deploys an external load balancer.
- External load balancer is deployed at Google Points of presence.
- Static IP for ingress lasts as long as the object.

To implement an internal Application Load Balancer, create an ingress object with the following settings:

- Routing depends on URL path, session affinity, and balancing mode of the backend NEGS.
- The object kind is ingress.
- Metadata requires an `Ingress.class: "gce-internal"` to spawn an internal load balancer.
- Proxies are deployed in a proxy only subnet in a specific region in your VPC.
- Only NEGs are supported. Use the following annotation in your service metadata:
 - `cloud.google.com/neg: '{"ingress": true}'`
- Forwarding rule is assigned from the GKE node address range.

4.2 | Diagnostic Question 04 Discussion



What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services
- D. Deployments

Question:

What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

4.2 | Diagnostic Question 04 Discussion



What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services
- D. Deployments



Feedback:

A. Pod templates

Feedback: Incorrect. Pod templates define how pods will be configured as part of a deployment.

B. Pods

Feedback: Incorrect. Pods provide the executable resources your containers run in.

*C. Services

Feedback: Correct! Service endpoints are defined by pods with labels that match those specified in the service configuration file. Services then specify how those pods are exposed.

D. Deployments

Feedback: Incorrect. Deployments help you with availability and the health of a set of pod replicas. They do not help you configure external access.

Where to look:

<https://cloud.google.com/kubernetes-engine/docs/concepts/kubernetes-engine-overview>

<https://cloud.google.com/kubernetes-engine/docs/concepts/pod>

<https://cloud.google.com/kubernetes-engine/docs/concepts/deployment>

<https://cloud.google.com/kubernetes-engine/docs/concepts/service>

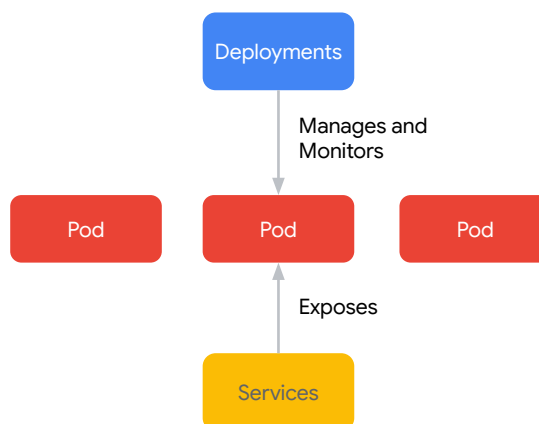
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M5 Containers in the Cloud
- Getting Started with Google Kubernetes Engine (ILT and On-demand)
 - M3 Kubernetes Architecture
- Skill Badge: Develop your Google Cloud Network
(https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Kubernetes objects



What is a pod? A pod is the smallest deployable object in Kubernetes. It is a single instance of a running process that contains one or more Docker containers. Pods provide networking and storage to containers, and contain dependencies the container needs to run and communicate.

What is a deployment? A deployment manages a set of multiple identical pods. It uses a replica set to define the number of pods. A deployment monitors pods in a replica set and replaces unhealthy instances to ensure your application remains available. A deployment uses a pod template, which provides a spec of what each deployed pod should look like. When you update the pod template in a deployment, it starts a rolling upgrade of the pods in the deployment.

What is a service? A service is a group of pod endpoints that you can configure access for. You use selectors to define which pods are included in a service. A service gives you a stable IP that belongs to the service. Pods have internal IP addresses but they can change as pods get restarted and replaced. A service can be configured to implement load balancing.

4.2 | Diagnostic Question 05 Discussion



What is the declarative way to initialize and update Kubernetes objects?

- A. kubectl apply
- B. kubectl create
- C. kubectl replace
- D. kubectl run

Question:

What is the declarative way to initialize and update Kubernetes objects?

4.2 | Diagnostic Question 05 Discussion



What is the declarative way to initialize and update Kubernetes objects?

A. **kubectl apply**



B. kubectl create

C. kubectl replace

D. kubectl run

Feedback:

*A. **kubectl apply**

Feedback: Correct! **kubectl apply** creates and updates Kubernetes objects in a declarative way from manifest files.

B. **kubectl create**

Feedback: Incorrect. **kubectl create** creates objects in an imperative way. You can build an object from a manifest but you can't change it after the fact. You will get an error.

C. **kubectl replace**

Feedback: Incorrect. **kubectl replace** downloads the current copy of the spec and lets you change it. The command replaces the object with a new one based on the spec you provide.

D. **kubectl run**

Feedback: Incorrect. **kubectl run** creates a Kubernetes object in an imperative way using arguments you specify on the command line.

Where to look:

https://cloud.google.com/kubernetes-engine/docs/how-to/deploying-workloads-overview#imperative_commands
<https://kubernetes.io/docs/concepts/overview/working-with-objects/object-management/>

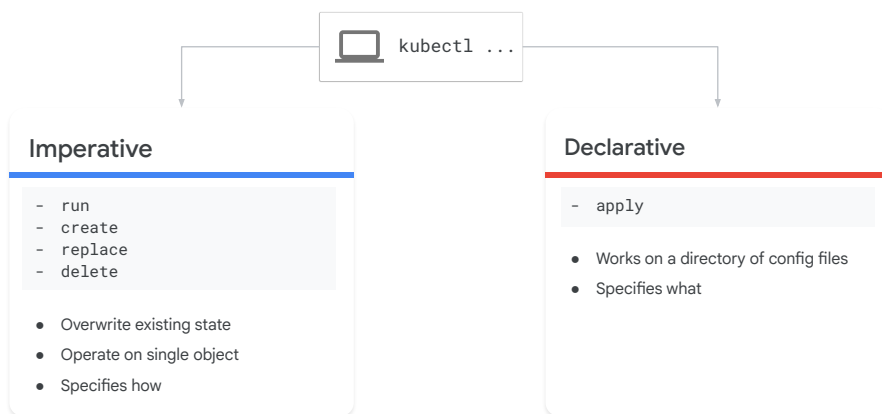
Content mapping:

- Getting Started with Google Kubernetes Engine (ILT and On-demand)
 - M4 Kubernetes Operations
- Skill Badge: Develop your Google Cloud Network (https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Types of kubectl commands



You execute kubectl commands to manage objects such as pods, deployments, and services.

Imperative commands such as run, create, replace, delete act on a live object or single config file and overwrite any state changes that have occurred on an existing object.

Declarative commands use a config stored in a directory to deploy and apply changes to your app objects.

- uses `kubectl -apply` on a directory
- You don't specify create, replace or delete commands.

Example commands:

<code>kubectl -run</code>	#generates a new object in a cluster, by default
of deployment	
<code>Kubectl -create</code>	#generates a new object from a config file
<code>Kubectl -get</code>	#display requested resources
<code>kubectl -expose</code>	#creates a new service that distributes traffic to
labelled pods	

Pods are not created by themselves but are based on template made available in deployments.

You can use the name of an existing object or defined config file. The object you create service for can be one of the following: deployment, service, replica set, replication controller or pod.

Important specs in a deployment manifest:

- kind:Deployment
- Replicas:3
- Spec:template specifies labels, container name, and image it is based on.

If you need to change a deployment you change the config file and do a `kubectl -apply`

Important specs in a service manifest:

Kind: service

Spec:selector - labels of pods included in service - have to match them all

Port: incoming port

targetPort: port the pod is listening on

4.2 Managing Google Kubernetes Engine resources

Courses

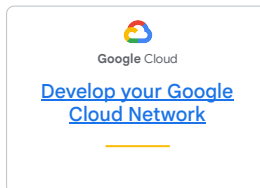
[Google Cloud Fundamentals: Core Infrastructure](#)

- M5 Containers in the Cloud

[Getting Started with Google Kubernetes Engine](#)

- M3 Kubernetes Architecture
- M4 Kubernetes Operations

Skill Badge



Documentation

[Ingress for internal Application Load Balancers](#)

[Ingress for external Application Load Balancers](#)

[Configure Ingress for external Application Load Balancers](#)

[Configuring Ingress for internal Application Load Balancers](#)

[GKE overview | Kubernetes Engine Documentation](#)

[Pod | Kubernetes Engine Documentation](#)

[Deployment | Kubernetes Engine Documentation](#)

[Services | Kubernetes Engine Documentation](#)

[Overview of deploying workloads | Kubernetes Engine Documentation](#)

[Kubernetes Object Management](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic questions we just reviewed are covered in these modules, skill badges, and documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

[Getting Started with Google Kubernetes Engine \(On-demand\)](#)

[Develop your Google Cloud Network \(Skill Badge\)](#)

<https://cloud.google.com/kubernetes-engine/docs/concepts/ingress-ilb>

<https://cloud.google.com/kubernetes-engine/docs/concepts/ingress-xlb>

<https://cloud.google.com/kubernetes-engine/docs/how-to/load-balance-ingress>

<https://cloud.google.com/kubernetes-engine/docs/how-to/internal-load-balance-ingress>

<https://cloud.google.com/kubernetes-engine/docs/concepts/kubernetes-engine-overview>

<https://cloud.google.com/kubernetes-engine/docs/concepts/pod>

<https://cloud.google.com/kubernetes-engine/docs/concepts/deployment>

<https://cloud.google.com/kubernetes-engine/docs/concepts/service>

https://cloud.google.com/kubernetes-engine/docs/how-to/deploying-workloads-overview#imperative_commands

<https://kubernetes.io/docs/concepts/overview/working-with-objects/object-management/>

4.3 | Managing Cloud Run resources

Considerations include:

- Deploying new versions of an application
- Adjusting application traffic splitting parameters
- Setting scaling parameters for autoscaling instances

Google Cloud

Cloud Run and Cloud Run functions are Google's serverless approach to handling containers and functional code. In both of these technologies, you pay for resources based on how many requests are coming in. One big difference between the two of them is that Cloud Run is optimized for multiple concurrent connections to each instance, while Cloud Run functions only lets you have one connection per function instance.

For Cymbal Superstore, Cloud Run could be used to quickly test updates to containers. As an Associate Cloud Engineer, you could be tasked to implement traffic splitting to test changes or rollback updates that didn't work well. There are also settings you need to know for autoscaling, such as min and max instances, that will let you make tradeoffs of relative latency versus cost.

These types of tasks were covered in this question:

Question 7: Express the differences between manual, basic, and automatic scaling in serverless: App Engine or Cloud Run

4.3 | Diagnostic Question 06 Discussion



You have a Cloud Run service with a database backend. You want to limit the number of connections to your database.

What should you do?

- A. Set Min instances.
- B. Set Max instances.
- C. Set CPU Utilization.
- D. Set Concurrency settings.

Question:

You have a Cloud Run service with a database backend. You want to limit the number of connections to your database. What should you do?

4.3 Diagnostic Question 06 Discussion



You have a Cloud Run service with a database backend. You want to limit the number of connections to your database.

What should you do?

- A. Set Min instances.
- B. Set Max instances.
- C. Set CPU Utilization.
- D. Set Concurrency settings.



Google Cloud

Feedback:

A. Set Min instances.

Feedback: Incorrect. Min instances reduce latency when you start getting requests after a period of no activity. It keeps you from scaling down to zero.

*B. Set Max instances.

Feedback: Correct! Max instances control costs, keeping you from starting too many instances by limiting your number of connections to a backing service.

C. Set CPU Utilization.

Feedback: Incorrect. Default CPU utilization is 60%. It doesn't affect the number of connections to your backing service.

D. Set Concurrency settings.

Feedback: Incorrect. Concurrency is how many users can connect to a particular instance. It does not directly affect connections to backend services.

Where to look:

<https://cloud.google.com/run/docs/about-instance-autoscaling>

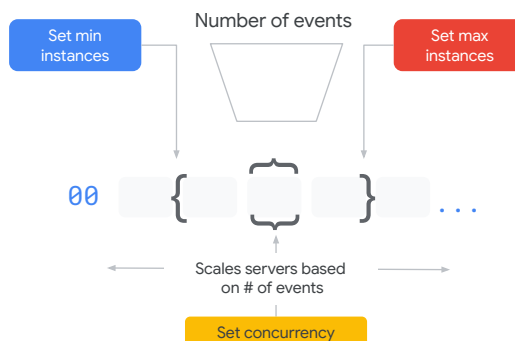
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M6 Applications in the Cloud

Summary:

Explanation/summary on the following slide.

Cloud Run autoscaling



How does autoscaling work in Cloud Run?

Cloud Run automatically scales the number of container instances required for each deployed revision. When no traffic is received, the deployment automatically scales to zero.

Other ways you can affect Cloud Run autoscaling:

- CPU utilization, with a default 60% utilization.
- Concurrency settings, with a default of 80 concurrent requests. You can increase it to 1000. You can also lower it if you need to.
- Max number of instances limits total number of instances. It can help you control costs and limit connections to a backing service. Defaults to 1000. Quota increase required if you want more.
- Min number of instances keeps a certain number of instances up. You will incur cost even when no instances are handling requests.

Instances that are started might remain idle for up to 15 minutes to reduce latency associated with cold starts. You don't get charged for these idle instances. You set a min and max on the container tab in the advanced settings dialog.

4.3 | Managing Cloud Run resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M6 Applications in the Cloud

Documentation

[About container instance autoscaling | Cloud Run Documentation](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic question we just reviewed are covered in this modules and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

<https://cloud.google.com/run/docs/about-instance-autoscaling>

4.4 | Managing storage and database solutions

Considerations include:

- Managing and securing objects in Cloud Storage buckets
- Setting object lifecycle management policies for Cloud Storage buckets
- Executing queries to retrieve data from data instances (e.g., Cloud SQL, BigQuery, Spanner, Firestore, AlloyDB)
- Estimating costs of data storage resources
- Backing up and restoring database instances (e.g., Cloud SQL, Firestore)
- Reviewing job status (e.g., Dataflow, BigQuery)

Google Cloud

We discussed earlier about how to define an external table definition in BigQuery. Cymbal Superstore's transportation management app required you to do this so that you could query the location of Cymbal Superstore's delivery vehicles from Bigtable. That is just one example of the type of storage management tasks you will have to do as an Associate Cloud Engineer

Consider this example. You store static images of products for Cymbal Superstore's ecommerce app in Cloud Storage. As an Associate Cloud Engineer, you would be expected to know how to secure access to these images from the application through IAM roles assigned to a service account. When you upgrade product images you would like to keep the previous images, but move them to a different storage type based on object versioning. You could do this using the object lifecycle management feature of Cloud Storage.

You explored these types of tasks in this question:

Question 8: Implement different types of Google Cloud Storage Lifecycle Actions (delete, set storage class) using lifecycle conditions.

4.4 | Diagnostic Question 07 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

- A. Age
- B. CreatedBefore
- C. MatchesStorageClass
- D. IsLive
- E. NumberofNewerVersions

Question:

You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date. What conditions should you use? (Pick two.)

4.4 | Diagnostic Question 07 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

- A. Age
- B. CreatedBefore
- C. MatchesStorageClass
- D. IsLive
- E. NumberofNewerVersions



Feedback:

A. Age

Feedback: Incorrect. Age is specified by number of days, not a specific date.

*B. CreatedBefore

Feedback: Correct! CreatedBefore lets you specify a date.

*C. MatchesStorageClass

Feedback: Correct! MatchesStorageClass is required to look for objects with a Standard storage type.

D. IsLive

Feedback: Incorrect. IsLive has to do with whether or not the object you are looking at is the latest version. It is not date-based.

E. NumberofNewerVersions

Feedback: Incorrect. NumberofNewerVersions is based on object versioning and you don't specify a date.

Where to look:

<https://cloud.google.com/storage/docs/lifecycle>

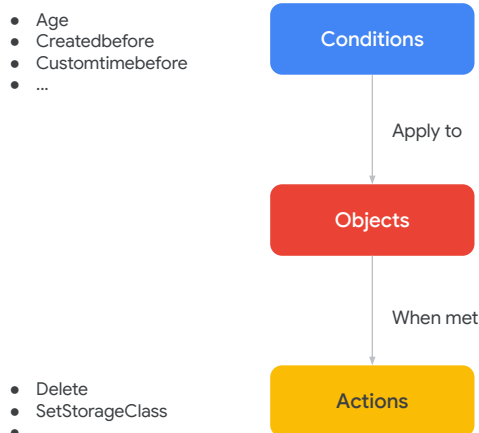
Content mapping:

- Google Cloud Fundamentals: Core Infrastructure (ILT and On-demand)
 - M4 Storage in the Cloud
- Architecting with Google Compute Engine (ILT)
 - M5 Storage and Database Services
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M2 Storage and Database Services

Summary:

Explanation/summary on the following slide.

Cloud Storage Lifecycle Actions



Lifecycle management configurations apply to current and future objects in a Cloud Storage bucket. When object metadata meets the criteria of any of the rules, Cloud Storage performs a specified action.

Examples:

- Downgrade the storage class of objects older than 365 days to coldline storage.
- Delete an object that existed before a certain date.
- Keep the 3 most recent versions of each object in a bucket with versioning enabled.

Object metadata has to match all rules for the action to fire. If an object state matches more than one rule set, delete takes precedence, followed by storage class with lowest price.

Here are the available conditions:

- Age
- Createdbefore
- Customtimebefore
- Dayssincecustomtime
- Dayssincenoncurrent
- Islive
- Matchesstorageclass
- Noncurrenttimebefore

- numberofnewerversions

4.4 Managing storage and database solutions

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M4 Storage in the Cloud

[Architecting with Google Compute Engine](#)

- M5 Storage and Database Services



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[Essential Google Cloud Infrastructure: Core Services](#)

- M2 Storage and Database Services



Documentation

[Object Lifecycle Management | Cloud Storage](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic question we just reviewed are covered in this module and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Google Cloud Fundamentals: Core Infrastructure \(On-demand\)](#)

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Core Services \(On-demand\)](#)

<https://cloud.google.com/storage/docs/lifecycle>

4.5 | Managing networking resources

Considerations include:

- Adding a subnet to an existing VPC
- Expanding a subnet to have more IP addresses
- Reserving static external or internal IP addresses
- Working with Cloud DNS and Cloud NAT

Google Cloud

As an Associate Cloud Engineer, any app you deploy is going to have connectivity requirements. Google's software defined networking stack is based on the idea of a Virtual Private Cloud. VPCs group regional resources into internal IP address ranges called subnets. As you manage network resources, you might have to add subnets or expand a subnet to let it support more devices. IP addresses assigned to both internal and external virtual machines are ephemeral, meaning as resources come and go your IP addresses might change. To get around this problem you can set and attach static IPs that persist across different individual resources.

How do these tasks apply to Cymbal Superstore? The ecommerce app requires global external connectivity for users to access it. You can manage this through an ingress object in GKE. The ecommerce application middleware is going to need private, regional, internal access to a Spanner backend that stores order data. Cymbal's supply chain app is going to need external regional connectivity with regional internal connectivity as well, only implemented with Google Cloud load balancers instead of GKE ingress objects.

You explored these types of tasks in the following questions:

Question 9: Describe how to expand the IPs available to a subnet (e.g. shrinking the subnet mask)

4.5 | Diagnostic Question 08 Discussion



Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 users or devices.

What should you do?

- A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`
- B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- C. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 22`

Question:

Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 new users or devices. What should you do?

4.5 | Diagnostic Question 08 Discussion



Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 users or devices.

What should you do?

- A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`
- B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- C. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21` ✓
- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 22`

Google Cloud

Feedback:

A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`

Feedback: Incorrect. A prefix length of 20 would expand the IP range to 4094, which is far too many for the scenario.

B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`

Feedback: Incorrect. This command is missing the compute command-set.

***C. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`**

Feedback: Correct! This command gives a total of 2046 addresses available and meets the requirement.

D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 22`

Feedback: Incorrect. This command doesn't give you enough IP addresses (only 1,000).

Where to look:

<https://cloud.google.com/sdk/gcloud/reference/compute/networks/subnets/expand-ip-range>

<https://cloud.google.com/vpc/docs/using-vpc#expand-subnet>

Content mapping:

- Architecting with Google Compute Engine (ILT)
 - M2 Virtual Networks
- Essential Google Cloud Infrastructure: Foundation (On-demand)
 - M2 Virtual Networks

Summary:

Explanation/summary on the following slide.

Expand IP addresses in a **subnet**

Current IP address range

Reduce your mask:
e.g. 24 to 20

Expanded address range

Command syntax:

```
gcloud compute networks subnets expand-ip-range SUBNET  
--prefix-length = PREFIX_LENGTH
```

If 10.0.128.0/24 you can supply 20 to reduce your mask, which increases your number of available ip addresses.

4.5 | Managing networking resources

Courses

[Architecting with Google Compute Engine](#)

- M2 Virtual Networks



[Essential Google Cloud Infrastructure: Foundation](#)

- M2 Virtual Networks



Documentation

[gcloud compute networks subnets expand-ip-range](#)
[Using VPC networks](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic question we just reviewed are covered in this module and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Foundation \(On-demand\)](#)

<https://cloud.google.com/sdk/gcloud/reference/compute/networks/subnets/expand-ip-range>

<https://cloud.google.com/vpc/docs/using-vpc#expand-subnet>

4.6 Monitoring and logging

Considerations include:

- Creating Cloud Monitoring alerts based on resource metrics
- Creating and ingesting Cloud Monitoring custom metrics (e.g., from applications or logs)
- Exporting logs to external systems (e.g., on-premises, BigQuery)
- Configuring log buckets, log analytics, and logs routers
- Viewing and filtering logs in Cloud Logging
- Viewing specific log message details in Cloud Logging
- Using cloud diagnostics to research an application issue
- Viewing Google Cloud status
- Configuring and deploying Ops Agent
- Deploying Managed Service for Prometheus
- Configuring audit logs

Google Cloud

Performance monitoring is another important aspect of managing the day-to-day operations of your cloud solutions. The majority of the products we review in this topic come from Google Cloud Observability.

For example, Cymbal Superstore's supply chain app might require you to monitor CPU utilization for all the instances in your managed instance group.

Cloud Monitoring allows you to build charts based on metrics you specify. You can also look at logs associated with resources from the dashboard. You could use this to monitor messages being posted to your pubsub topic for the transportation management app.

Custom metrics allow you to define metrics descriptors for things you want to keep track of that aren't included in standard metrics. For example, in Cymbal Superstore's ecommerce app you want to track the number of requests going to sales and the number going to support.

Cloud Logging allows you to log any timestamped data in logs you define and manage. There are a myriad of options for where you can save your logs and how to route them. There is also an interface provided to query your logs.

Cloud Observability has several tools that will help you with performance issues. Error Reporting, Cloud Trace, and Cloud Profiler all help you figure out what might be causing latency and performance issues in your apps.

You explored these types of tasks in this question:

Question 10: Configure a Google Cloud Operations custom alert: specify conditions, send optional notifications, and reference documentation.

4.6 Diagnostic Question 09 Discussion

Cymbal Superstore's supply chain management system has been deployed and is working well. You are tasked with monitoring the system's resources so you can react quickly to any problems. You want to ensure the CPU usage of each of your Compute Engine instances in us-central1 remains below 60%. You want an incident created if it exceeds this value for 5 minutes. You need to configure the proper alerting policy for this scenario.

What should you do?

- A. Choose resource type of VM instance and metric of CPU load, condition trigger if any time series violates, condition is below, threshold is .60, for 5 minutes.
- B. Choose resource type of VM instance and metric of CPU utilization, condition trigger all time series violates, condition is above, threshold is .60 for 5 minutes.
- C. Choose resource type of VM instance, and metric of CPU utilization, condition trigger if any time series violates, condition is below, threshold is .60 for 5 minutes.
- D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes.



Question:

Cymbal Superstore's supply chain management system has been deployed and is working well. You are tasked with monitoring the system's resources so you can react quickly to any problems. You want to ensure the CPU usage of each of your Compute Engine instances in us-central1 remains below 60%. You want an incident created if it exceeds this value for 5 minutes. You need to configure the proper alerting policy for this scenario. What should you do?

4.6 Diagnostic Question 09 Discussion

Cymbal Superstore's supply chain management system has been deployed and is working well. You are tasked with monitoring the system's resources so you can react quickly to any problems. You want to ensure the CPU usage of each of your Compute Engine instances in us-central1 remains below 60%. You want an incident created if it exceeds this value for 5 minutes. You need to configure the proper alerting policy for this scenario.

What should you do?

- A. Choose resource type of VM instance and metric of CPU load, condition trigger if any time series violates, condition is below, threshold is .60, for 5 minutes.
- B. Choose resource type of VM instance and metric of CPU utilization, condition trigger all time series violates, condition is above, threshold is .60 for 5 minutes.
- C. Choose resource type of VM instance, and metric of CPU utilization, condition trigger if any time series violates, condition is below, threshold is .60 for 5 minutes.
- D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes. 



Google Cloud

Feedback:

A. Choose resource type of VM instance and metric of CPU load, condition trigger if any time series violates, condition is below, threshold is .60, for 5 minutes.

Feedback: Incorrect. CPU load is not a percentage, it is a number of processes.

B. Choose resource type of VM instance and metric of CPU utilization, condition trigger all time series violates, condition is above, threshold is .60 for 5 minutes.

Feedback: Incorrect. The trigger should be "each of your instances", not "all of your instances."

C. Choose resource type of VM instance, and metric of CPU utilization, condition trigger if any time series violates, condition is below, threshold is .60 for 5 minutes.

Feedback: Incorrect. The alert policy should record an incident when the CPU utilization exceeds a certain amount. The condition for this statement is below that, so it is wrong.

* D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes.

Feedback: Correct! All the values of this statement match the scenario.

Where to look:

<https://cloud.google.com/monitoring/alerts/using-alerting-ui>

<https://cloud.google.com/monitoring/alerts>

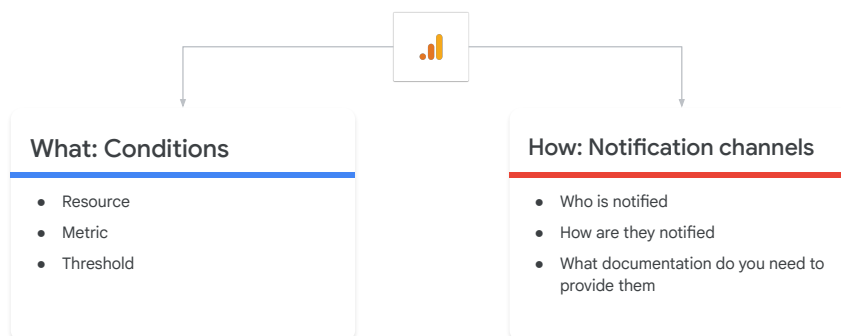
Content mapping:

- Architecting with Google Compute Engine (ILT)
 - M7 Resource Monitoring
- Essential Google Cloud Infrastructure: Core Services (On-demand)
 - M4 Resource Monitoring
- Skill Badges
 - Set Up an App Dev Environment on Google Cloud
(https://www.skills.google/course_templates/637)
 - Develop your Google Cloud Network
(https://www.skills.google/course_templates/625)

Summary:

Explanation/summary on the following slide.

Cloud operations custom alerts



In Cloud Monitoring you implement alerts by defining alert policies. An alerting policy specifies what you want to be alerted on and how you want to be notified. The “what” is made of conditions that describe the state of a resource, or groups of resources, that cause you to take action. The “how” is provided by notification channels, where you specify who is notified when the condition of the alerting policy is met. Each notification channel configures a different type of output, such as email, slack channel, or posting a message to pub/sub topic. You can also specify documentation you want included in your notification.

Conditions are made of a monitored resource, a metric for that resource, and the threshold where the condition is met. An alerting policy can have up to 6 conditions. In an alerting policy with 1 condition, when that condition is met, an incident is created. For an alerting policy, you specify how those conditions are combined.

4.6 | Monitoring and logging

Courses

[Architecting with Google Compute Engine](#)

- M7 Resource Monitoring



[Essential Google Cloud Infrastructure: Core Services](#)

- M4 Resource Monitoring



Skill Badges



[Set Up an App Dev Environment on Google Cloud](#)



[Develop your Google Cloud Network](#)

Documentation

[Managing metric-based alerting policies | Cloud Monitoring](#)

[Introduction to alerting | Cloud Monitoring](#)

Let's take a moment to consider resources that can help you build your knowledge and skills in this area.

The concepts in the diagnostic question we just reviewed are covered in this module and in this documentation. You'll find this list in your workbook so you can take a note of what you want to include later when you build your study plan. Based on your experience with the diagnostic questions, you may want to include some or all of these.

[Architecting with Google Compute Engine \(ILT\)](#)

[Essential Google Cloud Infrastructure: Core Services \(On-demand\)](#)

[Set Up an App Dev Environment on Google Cloud \(Skill Badge\)](#)

[Develop your Google Cloud Network \(Skill Badge\)](#)

<https://cloud.google.com/monitoring/alerts/using-alerting-ui>

<https://cloud.google.com/monitoring/alerts>